

LIZ MOTOR CORP Capital Budgeting for an ESG Project

A Financial Management Case Study



Project Background: The ESG Imperative

Liz Motor Corp., a leading US EV manufacturer, faces a critical challenge that necessitates a strategic investment in Environmental, Social, and Governance (ESG) initiatives.

1 Declining ESG Rankings

A recent 6.2% stock price drop followed the release of unfavorable 2022 ESG rankings, highlighting an urgent need for improvement.

2 Board Mandate

The Board has directed the CEO and CFO to significantly enhance ESG efforts to secure a quick turnaround in 2023 performance.

3 Strategic Project

This project aims to develop and install advanced solid-state batteries in most Liz electric vehicles.

4 Environmental Impact

A key objective is to reduce the carbon footprint of EV batteries by an impressive 39%, reinforcing our commitment to sustainability.



Project Financials: The Road Ahead

This ESG initiative requires a significant capital outlay, with initial investments totaling **\$800 million** in 2023 for technology acquisition. An additional **\$20 million** annually will be allocated from 2024 to 2027 for development and adoption. Projected cash inflows of **\$200 million** annually are expected from 2028 onwards, driven by increased sales and improved profit margins. The technology's lifespan is estimated at 12 years. The project will be funded via debt, with an after-tax cost of **8%**.

Year	Cash Flow
2023	-800
2024	-20
2025	-20
2026	-20
2027	-20
2028	200
2029	200
2030	200
2031	200
2032	200
2033	200
2034	200
2035	200

Key Benefits of the ESG Project



Reduced Capital Risk

Strong ESG performance fosters trust, stabilizes market perception, and lowers the cost of capital.



Enhanced Brand Value

Improves public perception and corporate image, boosting brand appeal and customer loyalty.



Competitive Differentiation

Leads to higher, more stable demand and reduced price sensitivity through unique product advantages.



Lower Operational Costs

Minimizes waste and promotes material reuse, efficiently cutting carbon-related expenses.



Mitigated Regulatory Risk

Aligns with sustainability standards, easing legal and compliance pressures significantly.



Access to Incentives

Increases eligibility for government green incentives, including enhanced EV tax credits.

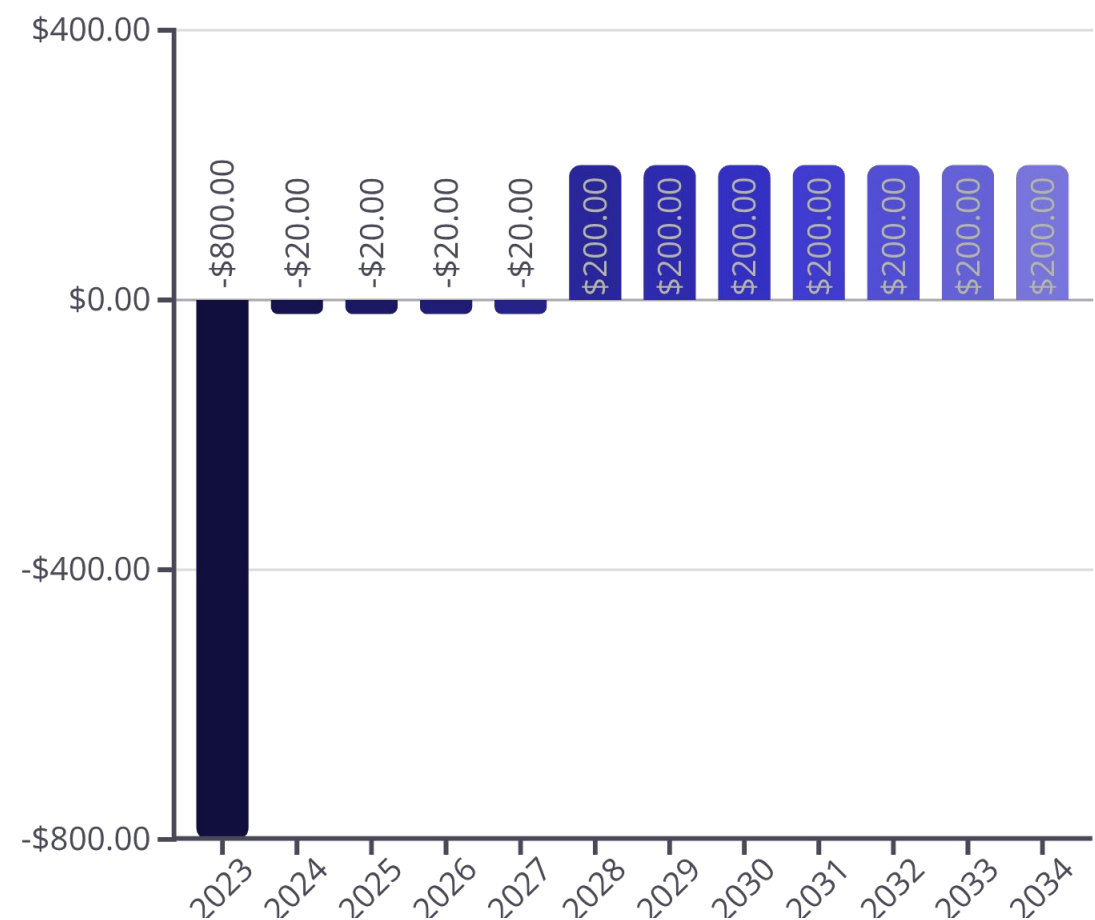


Improved Employee Metrics

Boosts job satisfaction and attracts top talent through purpose-driven work and a positive image.

Project Financial Analysis

To evaluate the ESG project's viability, a detailed financial analysis of the projected cash flows has been conducted. This includes initial investments, operating costs, and expected revenues over the project's lifespan, with a discount rate of 8%.



- 1

Initial Investment: \$800 million
- 2

Expected Life of the Project: 12 years

Operating Cost: \$20 million/year for first 4 years, \$0/year for next 8 years.
- 3

Annual Revenues

\$0/year for first 4 years, \$200 million/year for next 8 years
- 4

Discount Rate: 8%

This is same as after-tax cost of debt because financing is done through debt

This cash flow stream will be analyzed across 10 different scenarios to determine the project's Net Present Value (NPV), Internal Rate of Return (IRR), Profitability Index (PI), and Discounted Payback Period (DPP), ensuring a comprehensive understanding of its financial implications.

Key Financial Metrics: Definitions & Formulas

Understanding these metrics is crucial for evaluating the profitability and viability of the ESG project.

Net Present Value (NPV)

The difference between the present value of cash inflows and the present value of cash outflows over a period of time. A positive NPV indicates a profitable project.

$$NPV = \sum_{t=0}^n \frac{CF_t}{(1+r)^t}$$

Where: CF = Cash Flow, r = Discount Rate, t = Time Period

Internal Rate of Return (IRR)

The discount rate at which the Net Present Value (NPV) of all cash flows from a project or investment equals zero. It is used to evaluate the attractiveness of a project.

$$0 = \sum_{t=0}^n \frac{CF_t}{(1+IRR)^t}$$

IRR is calculated iteratively to find the rate where NPV is zero.

Profitability Index (PI)

A measure of a project's attractiveness. It is the ratio of the present value of future cash inflows to the initial investment. A PI greater than 1 suggests a worthwhile investment.

$$PI = \frac{\text{PV of Future Cash Inflows}}{\text{Initial Investment}}$$

Discounted Payback Period (DPP)

The length of time required for an investment's discounted cash flows to recover its initial cost. It considers the time value of money, making it a more accurate measure than simple payback period.

$$DPP = \text{Years before full recovery}$$

Effective IRR (with Inflation)

Adjusts the nominal IRR to reflect the real rate of return after accounting for inflation. This provides a more accurate picture of the investment's purchasing power.

$$\text{Real IRR} = \frac{(1 + \text{Nominal IRR})}{(1 + \text{Inflation Rate})} - 1$$

Project Financial Analysis: Base Case Scenario

Under the base case scenario, a detailed financial assessment of the ESG project reveals critical insights into its profitability and viability. The analysis includes discounted cash flows and key performance indicators, highlighting the project's financial outlook.

Discounted Cash Flows (Millions USD)

Year	Cash Flow	Present Value	Cumulative PV
2023	-800	-800.00	-800.00
2024	-20	-18.52	-818.52
2025	-20	-17.15	-835.67
2026	-20	-15.88	-851.54
2027	-20	-14.70	-866.24
2028	200	136.12	-730.13
2029	200	126.03	-604.10
2030	200	116.70	-487.40
2031	200	108.05	-379.35
2032	200	100.05	-279.30
2033	200	92.64	-186.66
2034	200	85.78	-100.88
2035	200	79.42	-21.46

Key Financial Metrics

-21.45

NPV

Million USD

0.97

Profitability Index

Lower than 1 indicates unprofitability.

7.6%

IRR

Below the 8% discount rate.

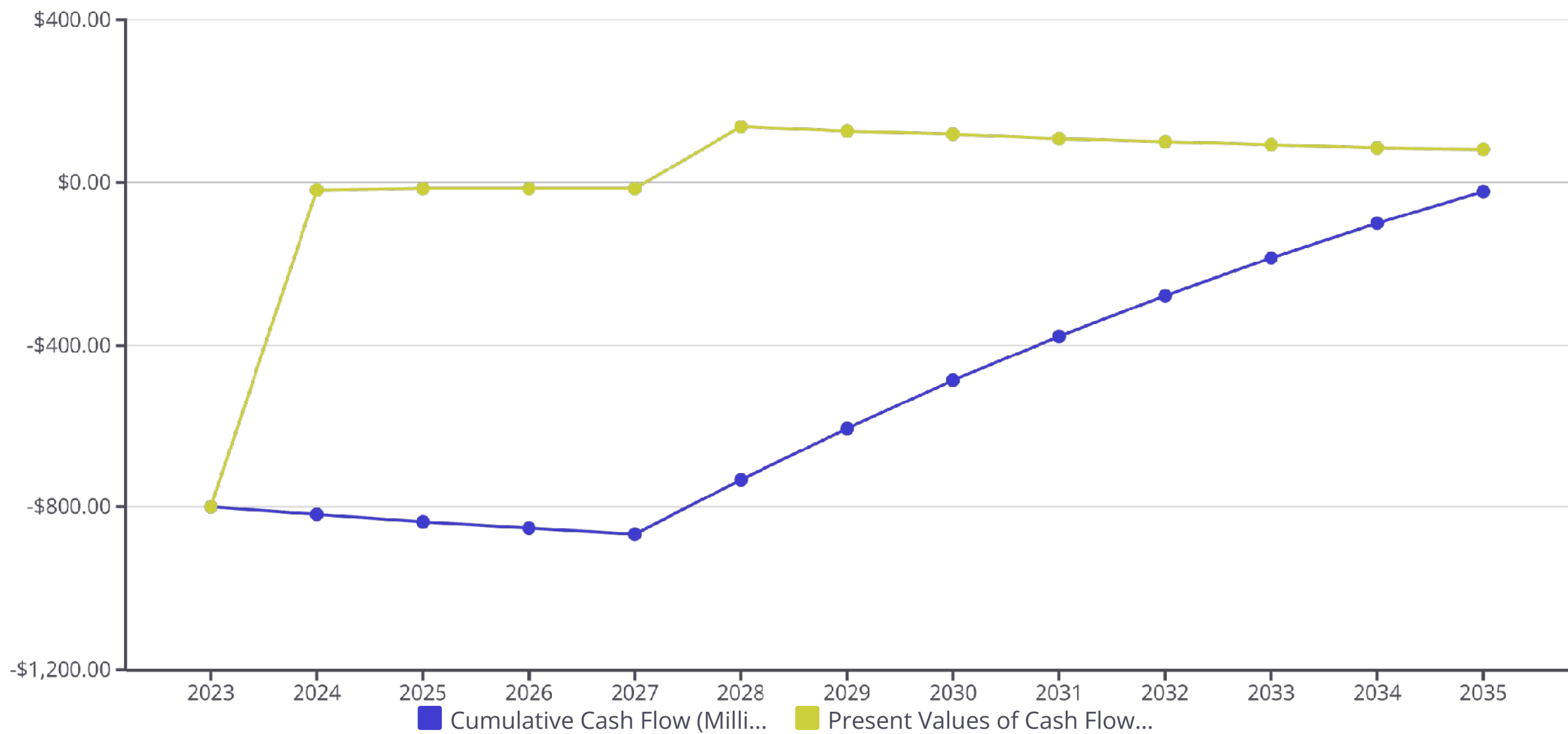
NIL

Discounted Payback Period

Project never recoups initial investment.

The initial analysis indicates that the project is **not financially profitable** under the current base case assumptions, as the Internal Rate of Return (IRR) of 7.6% falls below the 8% cost of debt, and the Net Present Value (NPV) is negative.

Cumulative Cash Flow Trend: Base case



The cumulative cash flow chart illustrates the project's inability to generate positive returns over its lifespan, maintaining a negative balance through 2035. This visual confirms the challenging financial viability of the ESG initiative under the current model.

Project Financial Analysis: Scenario 1 - Increased Interest Rates

In this scenario, we assess the ESG project's financial viability under a higher interest rate environment, where the after-tax cost of debt increases to 8.5%. This analysis projects the impact on key financial metrics.

Discounted Cash Flows (Millions USD)

Year	Cash Flow	Present Value	Cumulative PV
2023	-800	-800.00	-800.00
2024	-20	-18.43	-818.43
2025	-20	-16.99	-835.42
2026	-20	-15.66	-851.08
2027	-20	-14.43	-865.51
2028	200	133.01	-732.50
2029	200	122.59	-609.91
2030	200	112.99	-496.93
2031	200	104.13	-392.79
2032	200	95.98	-296.82
2033	200	88.46	-208.36
2034	200	81.53	-126.83
2035	200	75.14	-51.69

Key Financial Metrics

-51.69 NPV Million USD	0.94 Profitability Index Indicates unprofitability.
7.6% IRR Below the 8.5% discount rate.	NIL Discounted Payback Period Project never recoups initial investment.

Under Scenario 1, the project's financial outlook remains challenging. The Negative Net Present Value (NPV) of -\$51.69 million and an Internal Rate of Return (IRR) of 7.6% (below the 8.5% cost of debt) indicate that the project is **not financially viable** with increased interest rates.

Project Financial Analysis: Scenario 2 - Increased Green Bond Investment

This scenario explores the ESG project's financial viability if Liz Motor Corp. issues a new 12-year green bond, attracting socially responsible investors who accept a lower after-tax cost of debt of 7.5%.

Discounted Cash Flows (Millions USD)

Year	Cash Flow	Present Value	Cumulative PV
2023	-800	-800.00	-800.00
2024	-20	-18.60	-818.60
2025	-20	-17.31	-835.91
2026	-20	-16.10	-852.01
2027	-20	-14.98	-866.99
2028	200	139.31	-727.68
2029	200	129.59	-598.09
2030	200	120.55	-477.54
2031	200	112.14	-365.40
2032	200	104.32	-261.08
2033	200	97.04	-164.04
2034	200	90.27	-73.77
2035	200	83.97	10.20

Key Financial Metrics

10.20	1.013
NPV	Profitability Index
Million USD	Positive return on investment.
7.6%	12
IRR	Discounted Payback Period
Above the 7.5% discount rate.	Years

Under Scenario 2, the project's financial viability significantly improves. The positive Net Present Value (NPV) of \$10.20 million and an Internal Rate of Return (IRR) of 7.6% (exceeding the 7.5% cost of debt) indicate that the project is now **financially viable**.

Project Financial Analysis: Scenario 3 - CBAM Implementation

This scenario examines the project's financial outlook under the European Union's Carbon Border Adjustment Mechanism (CBAM) implementation in 2026, which is expected to significantly increase revenue from 2028 onwards due to carbon cost internalization.

Discounted Cash Flows (Millions USD)

Year	Cash Flow	Present Value	Cumulative PV
2023	-800	-800.00	-800.00
2024	-20	-18.52	-818.52
2025	-20	-17.15	-835.67
2026	-20	-15.88	-851.54
2027	-20	-14.70	-866.24
2028	220	149.73	-716.51
2029	220	138.64	-577.88
2030	220	128.37	-449.51
2031	220	118.86	-330.65
2032	220	110.05	-220.60
2033	220	101.90	-118.70
2034	220	94.35	-24.35
2035	220	87.37	63.02

Key Financial Metrics

63.03

NPV

Million USD

1.08

Profitability Index

Positive return on investment.

8.96%

IRR

Above the 8% discount rate.

12

Discounted Payback Period

Years

Under Scenario 3, the project demonstrates strong financial viability with a positive Net Present Value (NPV) of \$63.03 million and an Internal Rate of Return (IRR) of 8.96%, **exceeding even the 8.5% cost of debt**. This scenario significantly **improves the project's profitability and resilience**.

Project Financial Analysis: Scenario 4 - Technology Sale

This scenario considers a strategic exit for the ESG project: the sale of the developed technology to another automaker by the end of the 12th year (2035) for \$100 million. This introduces a significant salvage value, impacting the project's overall financial viability.

Discounted Cash Flows (Millions USD)

Year	Cash Flow	Present Value	Cumulative PV
2023	-800	-800.00	-800.00
2024	-20	-18.52	-818.52
2025	-20	-17.15	-835.67
2026	-20	-15.88	-851.54
2027	-20	-14.70	-866.24
2028	200	136.12	-730.13
2029	200	126.03	-604.10
2030	200	116.70	-487.40
2031	200	108.05	-379.35
2032	200	100.05	-279.30
2033	200	92.64	-186.66
2034	200	85.78	-100.88
2035	200 + 100(salvage)	79.42 + 39.71	18.31

Key Financial Metrics

18.31

NPV

Million USD

1.023

Profitability Index

Positive return on investment.

8.3%

IRR

Above the 8% discount rate.

12

Discounted Payback Period

Years

In Scenario 4, the inclusion of a technology sale at project end significantly improves the financial outlook. With a positive Net Present Value (NPV) of \$18.31 million and an Internal Rate of Return (IRR) of 8.3%, the project becomes **financially viable at the base 8% cost of debt, though it would not be at an 8.5% cost of debt.** The salvage value alone is a present value of \$39.71 million.

Project Financial Analysis: Scenario 5 - Enhanced Technology Value

This scenario models an optimistic outcome where the ESG project's developed technology proves significantly more valuable, generating \$210 million in annual cash flow from the fifth year (2028) and having an initial market value of \$810 million in 2023. Additionally, a salvage value of \$39.71 million (present value) is realized at the end of 2035 if the technology is sold.

Discounted Cash Flows (Millions USD)

Year	Cash Flow		Present Value	Cumulative PV
2023	-810		-810.00	-810.00
2024	-20		-18.52	-828.52
2025	-20		-17.15	-845.67
2026	-20		-15.88	-861.54
2027	-20		-14.70	-876.24
2028	210		142.92	-733.32
2029	210		132.34	-600.98
2030	210		122.53	-478.45
2031	210		113.46	-364.99
2032	210		105.05	-259.94
2033	210		97.27	-162.67
2034	210		90.07	-72.61
2035	210 + (salvage)	39.71	83.39 + 39.71	50.49

Key Financial Metrics

50.49	1.06
NPV	Profitability Index
Million USD	Positive return on investment.
8.8%	12
IRR	Discounted Payback Period
Above the 8% discount rate.	Years

Under Scenario 5, the project showcases exceptional financial viability. With a significant positive Net Present Value (NPV) of \$50.49 million and a robust Internal Rate of Return (IRR) of 8.8%, the project comfortably exceeds the 8% cost of debt, **guaranteeing profitability even at a higher 8.5% cost of debt. This scenario highlights a highly attractive investment opportunity.**

Project Financial Analysis: Scenario 6 - Increased Competition

This scenario explores the financial impact of increased competition from other major automakers, who are expected to launch similar technology by 2027. This market entry will reduce Liz Motor Corp.'s annual returns from the ESG project to \$180 million starting in 2028.

Discounted Cash Flows (Millions USD)

Year	Cash Flow	Present Value	Cumulative PV
2023	-800	-800.00	-800.00
2024	-20	-18.52	-818.52
-20	-17.15	-835.67	-835.66
2026	-20	-15.88	-851.55
2027	-20	-14.70	-866.25
2028	180	122.50	-743.75
2029	180	113.43	-630.32
2030	180	105.03	-525.29
2031	180	97.25	-428.04
2032	180	90.04	-338.00
2033	180	83.37	-254.63
2034	180	77.20	-177.43
2035	180 + 39.71 (salvage)	71.48 + 39.71	-66.24

Key Financial Metrics

-66.22

NPV

Million USD

0.917

Profitability Index

Indicates loss on investment.

6.87%

IRR

Below the 8% discount rate.

NIL

Discounted Payback Period

Project does not break even.

Under Scenario 6, the project faces significant financial challenges. The Net Present Value (NPV) is negative at -\$66.22 million, and the Internal Rate of Return (IRR) is a mere 6.87%, falling below the 8% cost of debt. This indicates that the **project would not be profitable and would fail to recoup its initial investment, even at a lower 7.5% cost of debt.**

Project Financial Analysis: Scenario 7 - Discontinuation due to Competition

This scenario examines the financial implications if Liz Motor Corp. were to discontinue the ESG project in 2028, following an initial investment and the entry of a competitor into the market. At the point of discontinuation, a salvage value of \$500 million from selling the technology to another automaker is assumed.

Discounted Cash Flows (Millions USD)

Year	Cash Flow	Present Value	Cumulative PV
2023	-800	-800.00	-800.00
2024	-20	-18.52	-818.52
2025	-20	-17.15	-835.67
2026	-20	-15.88	-851.55
2027	-20	-14.70	-866.25
2028 (Discontinued)	500 (salvage)	367.52	-498.73

Key Financial Metrics

-498.73

NPV

Million USD

0.376

Profitability Index

Indicates significant loss.

<0%

IRR

Never profitable

NIL

Discounted Payback Period

Project does not break even.

In Scenario 7, discontinuing the project in 2028 results in a negative NPV of -\$498.73 million and an IRR less than 0%, despite a \$500 million salvage value. This highlights that **early discontinuation, even with salvage value, is a highly unfavorable** outcome compared to continued operation in other scenarios.

Project Financial Analysis: Scenario 8 - Gradual Market Acceptance

This scenario models a gradual market adoption of the ESG project's technology. Starting in 2028, the project is anticipated to generate \$152 million in annual cash flow, with a consistent 5% annual growth thereafter. An initial investment of \$800 million is assumed for 2023.

Discounted Cash Flows (Millions USD)

Year	Cash Flow	Present Value	Cumulative PV
2023	-800	-800.00	-800.00
2024	-20	-18.52	-818.52
2025	-20	-17.15	-835.67
2026	-20	-15.88	-851.55
2027	-20	-14.70	-866.25
2028	152.00	103.45	-762.80
2029	159.60	100.58	-662.22
2030	167.58	97.78	-564.44
2031	175.96	95.07	-469.37
2032	184.76	92.42	-376.95
2033	193.99	89.86	-287.09
2034	203.69	87.36	-199.73
2035	213.88 + 39.71 (salvage)	84.93 + 39.71	-75.09

Key Financial Metrics

-75.09 NPV Million USD	0.91 Profitability Index Indicates loss on investment.
6.75% IRR Below the 8% discount rate.	NIL Discounted Payback Period Project does not break even.

Under Scenario 8, with gradual market acceptance and a 5% annual growth in returns, the project faces significant financial headwinds. The Net Present Value (NPV) remains negative at -\$75.09 million, and the Internal Rate of Return (IRR) is 6.75%, falling short of the 8% cost of debt. **This outcome indicates that the project would not achieve profitability, even at a lower after-tax cost of debt of 7.5%, making it an unfavorable investment.**

Project Financial Analysis: Scenario 9 - Hydrogen Breakthrough

This scenario explores the financial impact of a technological breakthrough in hydrogen-powered vehicles, which could quickly dominate the market and diminish the value of solid-state battery technology. With a 30% probability, this would shorten the ESG project's lifespan to 10 years and completely eliminate any salvage value.

Discounted Cash Flows (Millions USD)

Year	Cash Flow	Present Value	Cumulative PV
2023	-800	-800.00	-800.00
2024	-20	-18.52	-818.52
2025	-20	-17.15	-835.67
2026	-20	-15.88	-851.55
2027	-20	-14.70	-866.25
2028	200	136.12	-730.13
2029	200	126.03	-604.10
2030	200	116.70	-487.40
2031	200	108.05	-379.35
2032	200	100.05	-279.30
2033	200	92.64	-186.66

Key Financial Metrics

-186.66

NPV

Million USD

0.77

Profitability Index

Indicates loss on investment.

4.4%

IRR

Below the 8% discount rate.

NIL

Discounted Payback Period

Project does not break even.

7.13%

Expected IRR

Under Scenario 9, the project's financial outlook is significantly negative, primarily due to a shortened lifespan and no salvage value. The Net Present Value (NPV) is a substantial -\$186.66 million, and the Internal Rate of Return (IRR) is a mere 4.4%, falling far below the 8% cost of debt.

-28.15

Expected NPV

Million USD

This outcome signals that the project **would not be profitable, even with a reduced after-tax cost of debt of 7.5%**, making it an **unviable investment** under these conditions.

Project Financial Analysis: Scenario 10 - Incorporate an expected inflation rate of 3 percent

This scenario incorporates an expected inflation rate of 3%, leading to a real discount rate of 4.85% (calculated as (1.08 / 1.03) - 1). This adjustment allows for a more accurate assessment of the project's profitability in real terms, rather than nominal values, providing a clearer picture of its long-term viability against a backdrop of inflation.

Discounted Cash Flows (Millions USD) - Real Terms

Year	Cash Flow (Nominal)	Present Value (Real)	Cumulative PV (Real)
2023	-800	-800.00	-800.00
2024	-20	-19.07	-819.07
2025	-20	-18.19	-837.26
2026	-20	-17.35	-854.61
2027	-20	-16.55	-871.16
2028	200	157.83	-713.33
2029	200	150.53	-562.80
2030	200	143.57	-419.23
2031	200	136.92	-282.31
2032	200	130.59	-151.72
2033	200	124.55	-27.17
2034	200	118.79	91.62
2035	200 + 39.71 (salvage)	113.29 + 39.71	204.91 + 39.71 = 244.62

Key Financial Metrics

244.62	1.306
NPV	Profitability Index
Million USD	Positive investment return.
8.58%	11
IRR	Discounted Payback Period
Exceeds the 8% nominal discount rate.	Years

When inflation is incorporated, the project's financial outlook significantly improves. The Net Present Value (NPV) becomes a positive \$244.62 million, and the Internal Rate of Return (IRR) rises to 8.58%, **surpassing even the 8.5% cost of debt**. This demonstrates that accounting for inflation accurately paints a profitable picture for the ESG project, highlighting the importance of considering **inflation** in financial analysis.

Project Financial Analysis: Best & Worst Case Scenarios

This analysis consolidates findings from various financial scenarios to identify the most favorable and unfavorable outcomes for the ESG project, highlighting the critical impact of external factors like market shifts, technological advancements, and economic conditions on project viability.

Best Case: Scenario 3 - CBAM Implementation

Scenario 3, driven by Carbon Border Adjustment Mechanism (CBAM) implementation, emerges as the most robust financial outcome. The projected annual revenue of **\$220 million** significantly boosts profitability.

63.03

NPV

Million USD (with salvage)

1.079

Profitability Index

Strong positive return.

8.96%

IRR

Exceeds cost of debt.

12

Discounted Payback Period

Years

This scenario not only guarantees profit but also demonstrates resilience, maintaining viability even with an 8.5% after-tax cost of debt. Taking salvage and inflation into account would further increase the NPV and decrease the IRR.

The presence or absence of **salvage value** significantly impacts project viability. In Scenario 9, the complete loss of salvage value at the project's end drastically worsens the NPV. Conversely, scenarios retaining salvage value (like 8 and 10) provide a final cash inflow, cushioning the cumulative present value.

Furthermore, the incorporation of **inflation** (Scenario 10) reveals its critical role. By adjusting the nominal discount rate to a real discount rate, the project's profitability transformed from negative (in similar nominal scenarios) to a positive NPV of **\$244.62 million** and an IRR of **8.58%**. This emphasizes that assessing financial performance in real terms, rather than nominal, provides a more accurate and optimistic long-term view, reflecting the true future cash flows.

Worst Case: Scenario 9 - Hydrogen Breakthrough

Conversely, Scenario 9, anticipating a rapid hydrogen breakthrough, presents the most detrimental financial outlook. The project's shortened lifespan and complete loss of salvage value lead to substantial losses.

-186.66

NPV

Million USD

0.77

Profitability Index

Significant loss on investment.

4.4%

IRR

Well below cost of debt.

NIL

Discounted Payback Period

Project never breaks even.

This scenario underscores the risk of disruptive technologies rapidly rendering current investments obsolete. The project becomes unviable, highlighting the importance of continuous market monitoring and strategic planning to mitigate such risks.

Strategic Insights & Key Takeaways

This analysis yields several critical insights for future project planning and financial strategy, emphasizing the importance of detailed scenario assessment.

1 Leverage Green Bonds

Embrace green bond financing for ESG-aligned projects. This significantly reduces the after-tax cost of debt, thereby increasing the Net Present Value (NPV) and supporting environmental objectives.

3 Retain Core Technology

Avoid investing proprietary technology to competitors, as this leads to substantial financial losses and compromises long-term project value, as evidenced in Scenario 4.

2 Integrate Inflation Accurately

Crucially, incorporate inflation into financial models. Analyzing real cash flows provides a more optimistic and accurate long-term assessment of project viability, as demonstrated by improved NPV and IRR in Scenario 10.

4 Utilize Salvage Value

Plan to realize the salvage value of assets after the project's operational lifespan (e.g., after 12 years). This strategic approach adds significantly to the project's overall NPV.

Thank You

We appreciate your time and attention to our comprehensive financial analysis of the ESG project.